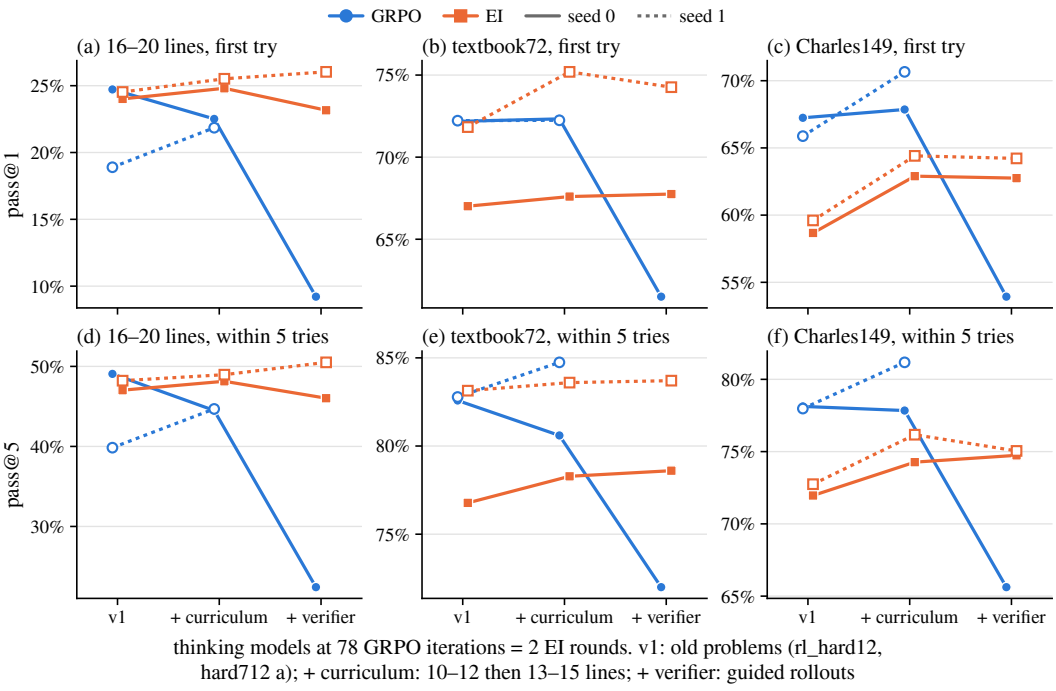
GRPO
EI
seed 0
seed 1
(a) 16–20 lines, first try
(b) textbook72, first try
(c) Charles149, first try
(d) 16–20 lines, within 5 tries
(e) textbook72, within 5 tries
(f) Charles149, within 5 tries
pass@1
pass@5
25%
20%
15%
10%
75%
70%
65%
70%
65%
60%
55%
50%
40%
30%
85%
80%
75%
80%
75%
70%
65%
v1
+ curriculum
+ verifier
thinking models at 78 GRPO iterations = 2 EI rounds. v1: old problems (rl_hard12, hard712 a); + curriculum: 10–12 then 13–15 lines; + verifier: guided rollouts